

BHARATH SIMHA

Senior Data Engineer

Ph: +1 437 247 5000 / gudursimha2224@gmail.com

Location: Mississauga, ON

Status: Canada Permanent Resident

Professional Expertise

- An accomplished **Database Professional** with 7+ years of experience as Data Engineer with expertise in SQL/PLSQL Strong understanding of Hadoop APIs like **HDFS and Map-Reduce, Spark, Hive, Pig, Sqoop, Impala, Oozie.**
- Experience with **Big Data, Python,** and **Scala** technologies for **data processing pipelines**, data ingestion, **cloud apps**, and **Dev-Ops.**
- Expert in **ingesting data** for incremental loads from various RBMS tools using Apache **Sqoop.**
- Using Apache **Kafka** developed scalable applications for real-time ingestions into various databases.
- Developed various scalable big data applications in **Azure HDInsight** for ETL services.
- Experience in using Azure cloud tools like **Azure data factory, Azure Data Lake, and Azure Synapsis.**
- Worked on building pipelines using **Snowflake** for extensive data aggregations.
- Experience with Python packages like **Pandas.**
- Experience in using various Hadoop Distributions like **Cloudera and Microsoft Azure.**
- Experience in building data pipelines using **Azure Data Factory, Azure data bricks, and loading data to Azure Data Lake, Azure SQL Database, Azure SQL Data warehouse,** and controlling and granting database access.
- Experience in Migrating SQL database to **Azure Data Lake, Azure data lake Analytics, Azure SQL Database, Data Bricks, and Azure SQL Data.**
- Migrating On-premises databases to **Azure Data Lake store** using **Azure Data factory.**
- Experience in moving data into and out of the HDFS and Relational Database Systems (RDBMS) using Apache Sqoop.
- Developed **Pig Latin** scripts and **MapReduce** jobs for large data transformations and Loads.
- Experience in building optimized ETL data pipelines using **Apache Hive and Spark.**
- Implemented various optimizing techniques in **Hive and Spark** scripts for data transformations.
- Experience in building ETL scripts in **Impala and Kudu** for faster access to reporting layer.
- Experience in NoSQL databases like **HBase and Cassandra and Mongo DB.**
- Ability to troubleshoot and tune relevant programming languages like **SQL, Python, Scala, PIG, Hive, RDDs, Data Frames & Map Reduce.**
- Replaced existing map-reduce jobs and **Hive scripts** with **Spark Data-Frame transformation and actions** for faster analysis of the data.
- Experienced in Extracting, Transforming and Loading **ETL data** from Excel, Flat file, Oracle to MS SQL Server by using **BCP utility, DTS and SSIS** services.
- Hands on experience in building queries to retrieve data into **Tableau** from SQL Server and Oracle server.
- Experienced in enhancing and **deploying** the **SSIS Packages** from **development** server to **production** server.
- Developed and maintaining different kinds of reports like Cascading, Drill-down, Drill-through, **dashboards** and **parameterized Reports** using SSRS. Managed security by defining roles and rights.
- Expertise gathering **functional specification** requirements for the **development** of **dashboards** and reports.

Technical Skills

Tools: Azure Data Factory, Azure Databricks, Azure Blob storage, Azure table storage, Amazon Web Services, Google Cloud Platform (GCP), AWS RedShift, WhereScape RED, Apache Hive, Apache Spark, Scala, MDM, Microsoft Office, SVN, Azure TFS, TFS, Octopus, Team City, Share point, MS Outlook, Tidal, Microsoft Service, Kubernetes, Docker, Snowflake.

SQL Server Tools: Azure SQL server, SQL Server Management Studio, SSIS, SSRS, SSAS, Data Visualization, Data Migration, Data Mining.

Operating System: Windows, Unix, Linux, MacOS

Packages: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn

Database: MS SQL Server 2017, 2016, 2012, Oracle, Snowflake.

Language: HTML, T-SQL, PL/SQL, UNIX, Java, C++, .Net, Python

Reporting: SSRS, Power BI, Tableau.

Experience

WestJet Airways, Canada

Nov 2023 to Present

Designation: Senior Data Engineer

- Led the development and implementation of **data warehouse** solutions using **Wherescape Red** and **Snowflake**, resulting in a 40% reduction in data processing time and increased scalability.
- Designed and maintained **ETL (Extract, Transform, Load) processes** within **Wherescape Red** for efficient data integration from diverse sources into Snowflake data warehouse.
- Leveraged **Snowflake's** capabilities for scalable and elastic data storage, processing, and analytics, ensuring high performance and reliability of **data solutions**.
- Collaborated closely with stakeholders to gather requirements, design **data models**, and develop **automated workflows** for seamless data ingestion and transformation.
- Optimized SQL queries and Snowflake's performance features for enhanced query performance and faster data retrieval, enabling timely insights delivery to business users.
- Migration from on prem **SQL Server DB** to Snowflake.
- Used Azure Data Lake as storage, **Azure Data Factory** for orchestration, **Azure Databricks, Spark, Python, SQL**
- Analyzed the architecture of the existing on Prem system and worked on migrating the same to Microsoft Azure
- Maintained system documentation including **flow diagrams**, use cases and **architecture diagrams**.
- Built data frameworks to manage deadlocks, recovery, and **logging pipeline data**; implemented delta logic extractions for multiple sources with the use of **control tables**.
- Using **Azure Data factory, T-SQL, Spark-SQL, and U-SQL** azure data lake in multiple formats like JSON, Parquet. Extract, transform and load data from sources systems to azure data storage services. Processing data in **Azure Data bricks** and ingesting it into one or more Azure services.
- Migration of Data from source systems and **Azure to Snowflake** using copy command, Snow pipe, tasks, procedures which can be used for downstream reporting requirement.
- Working experience Pipelines in ADF using Linked Services/Datasets/Pipeline to Extract and load data from different sources like **Azure SQL, ADLS, Blob storage, Azure SQL Data warehouse**.
- Working with Azure Data Factory Data transformations such as **Stored Procedure**, Databricks.
- Implemented architectures using Azure Data platform capabilities like Azure Data Lake, Azure Data Factory, HDInsight, **Azure SQL, Azure SQLDW**, and Azure SQL Managed Instances.
- Installed Self hosted IR in on premise machine and created the connection from ADF.
- Created Azure Data Factory V2 Linked Services to connect to different sources like on premise SQL server, blob storage and SQL Data Warehouse.

TELUS, Canada

April 2023 to October 2023

Designation: Senior Data Engineer

- Developed **Azure Data bricks** notebooks to apply the business transformations and perform data cleansing operations.
- Developed **Data bricks Python** notebooks to Join, filter, pre-aggregate, and process the files stored in **Azure data lake storage**.
- Created reusable pipelines in **Data Factory** to extract, transform and load data into **Azure SQL DB and SQL Data warehouse**.
- Knowledge of modern **cloud-based warehouse technologies** such as **BigQuery, Snowflake, Firebolt, and Amazon Redshift** is crucial for data engineers to choose the best possible warehouse architecture, set it up, and maintain it properly.
- **Design and implement** Modern enhancements, updates, and programming changes for portions and subsystems of **data pipelines, repositories, and models for structured/unstructured data**
- Implemented and **managed ETL solutions**, focusing on automating operational processes and ensuring high data quality.
- Developed **JSON scripts** for deploying the Pipeline in **Azure Data Factory (ADF)** that process the data using the SQL Activity.
- Develop **conceptual solutions** & create proof-of-concepts to demonstrate viability of solutions.
- Design Setup maintain Administrator the **Azure SQL Database, Azure Analysis Service, Azure SQL Data warehouse, Azure Data Factory, Azure SQL Data Warehouse**.
- Experienced in working as part of fast paced **Agile Teams**, exposure to testing in scrum team, Test-Driven development.
- Experience in publishing the **Power BI Desktop models** to Power Bi Service to create highly **informative dashboards**, collaborate using workspaces, apps, and to get quick insights about datasets.
- For the creation of **calculations** and **measures** in the Tabular Models used **DAX** ((Data Analysis Expressions).
- A team player with **excellent interpersonal, communication and presentation skills**.

Fiera Capital Corporation, Canada

March 2022 to March 2023

Designation: Senior Data Engineer

- Developed an automated process in **Azure cloud** that can ingest data daily from web service and load it to **Azure SQL DB**.
- Experience in building data pipelines using **Azure Data Factory, Azure data bricks, and loading data to Azure Data Lake, Azure SQL Database, Azure SQL Data warehouse**, and controlling and granting database access.
- Experience in Migrating SQL database to **Azure Data Lake, Azure data lake Analytics, Azure SQL Database, Data Bricks, and Azure SQL Data warehouse** and controlling and granting database access and Migrating On-premises databases to **Azure Data Lake** store using **Azure Data factory**.
- Extract Transform and Load data from Sources Systems to **Azure Data Storage services** using a combination of **Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics**. Data Ingestion to one or more Azure Services - (**Azure Data Lake, Azure Storage, Azure SQL, Azure DW**) and processing the data in In **Azure Databricks**.

- Implemented Proof of concepts for **SOAP & REST APIs**.
- Used Spark and **Spark-SQL** to read the parquet data and create the tables in hive using the Scala API.
- Expertized in Implementing Spark using **Scala** and **Spark SQL** for faster testing and processing of data responsible to manage Data from different sources.
- Responsible in development of **Spark Cassandra** connector to load data from flat file to Cassandra for analysis.
- Involved in Performance tuning of **ETL transformations, data validations** and **stored procedures**.
- Strong experience in designing and implementing ETL packages using SSIS for integrating data using OLE DB connection from **heterogeneous sources**.
- Created Complex ETL Packages using SSIS which upsets data from **staging table to database tables**.
- Experience in creating reports from scratch using **Power BI** from Excel workbook that contains Power View sheets.
- Created roles using **SSAS** to restrict cube properties.
- Implemented **cell level security** in cubes using **MDX** expressions to restrict users of one region seeing data of another region **using SSAS**.
- Created calculated measures using **MDX** implementing business requirement.
- Experienced working on Star and Snowflake Schemas and used the fact and dimension tables to build the cubes, perform processing and deployed them to **SSAS database**.
- Ingested data from data sources using a combination of **SQL, Google analytics API, and Salesforce API** using python to create data views to be used **in BI tools**
- Worked closely with **Quality Assurance Analyst and Business Analyst**.

Gore Mutual, Canada

Jan 2021 to Feb 2022

Designation: Senior Data Engineer

- Developed and maintained end-to-end operations of **ETL data pipelines** and worked with large data sets in **Azure Data Factory**.
- Created Pipelines in **ADF** using **Linked Services/Datasets/Pipeline/** to Extract, Transform, and load data from different sources like **Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool and backwards**.
- Worked closely with a global team of **Data Scientists, Domain experts, 10+ developers** and Product Managers to create business impacting data analytics services.
- Configured **Spark streaming** to get ongoing information from **Kafka** and store the stream information to HDFS.
- Extract Real time feed using **Kafka** and **Spark Streaming** and convert it to RDD and process Data in the form of Data Frame and save the data as parquet format in HDFS.
- Performed the role of **Scrum Master** in daily stand-up calls.
- Researched and implemented various components like pipeline, activity, mapping data flows, data sets, linked services, triggers, and control flow.
- Performed extensive debugging, **data validation, error handling mechanism**, transformation types and data clean up analysis within large datasets.
- Collaborated with DevOps Engineers to developed **automated CI/CD and Test-driven development** pipeline using **Azure** as per client standards.
- Collaborated with Analytics and BI teams to develop globally adopted metrics and reporting on-demand solutions, **reducing manual data analysis by over 50%**.

- Architected an automated environment using **PowerShell and Azure Cloud Shell** to deploy Azure data solutions, **driving cost savings of 25%**.
- Developed data models that streamlined data processing pipelines in the **Azure environment**, resulting in an **increase of 30% in productivity**.
- Upgraded SQL Server 2008 and 2008 R2 **Database into SQL Server 2012, 2014, and 2016**.
- Export the current Data Models into **PDF out of Erwin** and publish them on to **SharePoint for various users**.
- Developing, Administering, and Managing corresponding databases: **Consolidated Data Store, Reference**
- Writing Triggers, Stored Procedures, Functions, Coding using Transact-**SQL (TSQL)**, Create and maintain.

Boeing, Canada

Aug 2018 to Sep 2019

Designation: Data Engineer

- Effectively interacted with **Business Analysts and Data Modelers** and defined Mapping documents and Design process for various Sources and Targets.
- Gathered user **requirements, analyzed, and designed software solution** based on the requirements.
- Involved in creating database objects like tables, views, procedures, triggers, functions using **T-SQL** to provide definition, structure and to maintain data efficiently.
- Building, publishing customized interactive reports and dashboards, report scheduling using **Tableau server**.
- Created action filters, parameters, and calculated sets for preparing **dashboards and worksheets in Tableau**.
- Effectively used **data blending feature in tableau**.
- Defined best practices for **Tableau report** development.
- Administered user, user groups, and scheduled instances for reports in Tableau.
- **Executed and tested** required queries and reports before publishing.
- Converted metric insight reports to **tableau reports**.
- See a comprehensive view of your **sales pipeline and results**.
- Create, customize share interactive **web dashboards** in minutes with simple drag drop.

Flipkart, Hyderabad, India

Dec 2016 to Aug 2018

Designation: Database Developer

- Create, customize share interactive **web dashboards** in minutes with simple drag drop.
- Developed and maintained **Java-based applications**, contributing to the full software development lifecycle.
- Utilized **Java EE** to design and implement server-side components, ensuring efficient and scalable application architecture.
- Collaborated with frontend developers to integrate user-facing elements using **HTML, CSS, and JavaScript**.
- Implemented responsive and user-friendly **front-end interfaces** using **React and Angular**.
- Designed and developed **RESTful APIs** and microservices using **C#** and **Python Flask**.
- Integrated third-party APIs to enhance application functionality.
- Collaborated closely with **UI/UX designers** to ensure a seamless user experience.
- Utilized Git for version control and participated in code reviews to maintain code quality.

- Worked within an **Agile/Scrum development** environment to meet project milestones and deadlines.
 - Deployed applications on **AWS/Azure/Heroku** and optimized server configurations for performance.
 - Designed embedded systems software using concepts of engineering, computer science, and mathematical analysis.
 - Oversaw installation of software systems while collaborating on implementing new features.
-

Education

- B. Tech Computer Science, J.N.T.U Hyderabad, Telangana, India.
 - PGDM Information Technology Business Analysis, Conestoga College, ON, Canada.
-